Harness



File: harness.kicad_sch

Daughterboard



File: daughter.kicad_sch

LSU, MAP, IAT, USB, AUX



File: io.kicad_sch

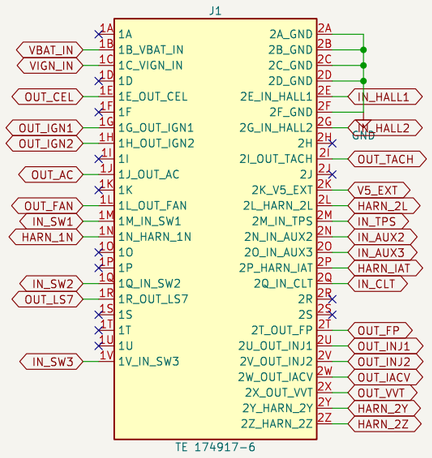
Miata ECU motherboard 1989–1993 (NA6). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

Sheet: /
File: miata_mb_8993.kicad_sch

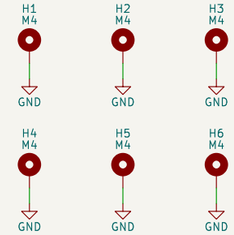
Title: Miata ECU motherboard 1989–1993 (NA6)

Size: A4	Date: 2026-10-30	**Rev: 0.11**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Stock harness header (Miata ECU motherboard 1989–1993): every pin labelled with the daughterboard signal it drives



Case mounting bosses (M4), tied to ground



Generated from scripts/mb_na6.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

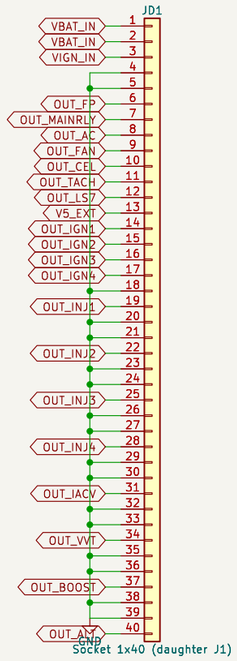
Title: Miata ECU motherboard 1989–1993 — Harness

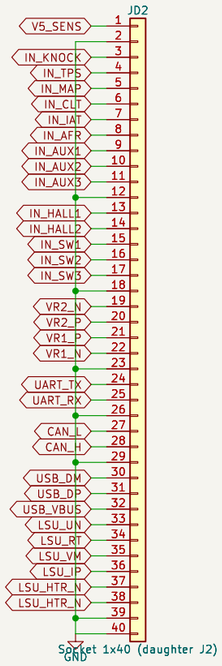
Size: A3	Date: 2026-10-30	**Rev: 0.11**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

Daughterboard standoffs (M3)

HD1 M3 standoff
HD2 M3 standoff
HD3 M3 standoff
HD4 M3 standoff





Generated from scripts/mb_na6.py; netlist identical to the routed board

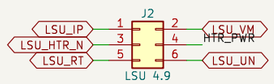
Sheet: /Daughterboard/
File: daughter.kicad_sch

Title: Miata ECU motherboard 1989–1993 — Daughterboard

Size: A3	Date: 2026-10-30	**Rev: 0.11**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

LSU 4.9 wideband sensor (controller on the daughterboard)

LSU 4.9: 1 IP 2 VM 3 H− 4 H+ 5 RT 6 UN

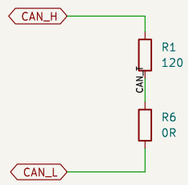




Heater fuse: 2 A hold, 30 V PTC

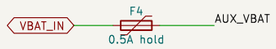
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

CAN bus termination at the ECU end

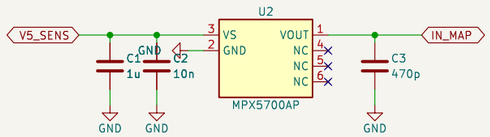


120R; remove R6 if the bus is terminated elsewhere

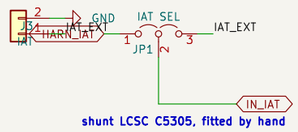
Fused +12 V feeds (harness actuators, AUX)



MAP sensor

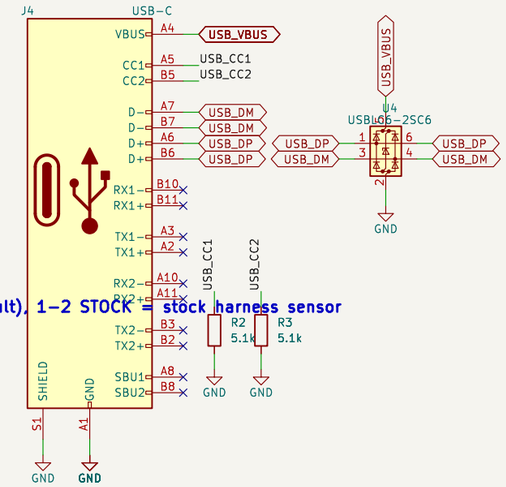


IAT source jumper JP1: 2–3 EXT = onboard connector J3 (default), 1–2 STOCK = stock harness sensor

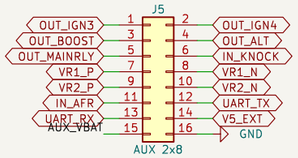


shunt LCSC C5305, fitted by hand

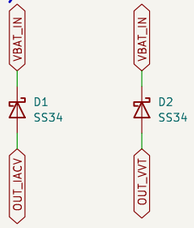
USB-C (device): tuning, logs, DFU firmware update



AUX: add-on I/O the stock harness has no pins for (+12 V through F4)

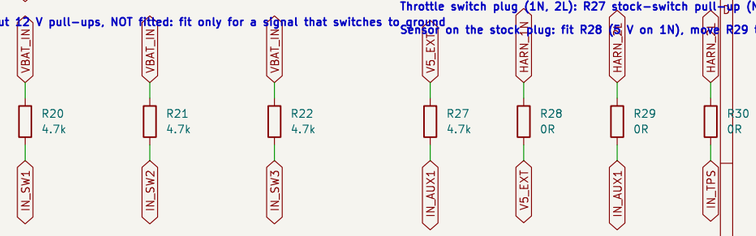


Valve freewheel diodes (to each valve's +12 V feed) and fit options



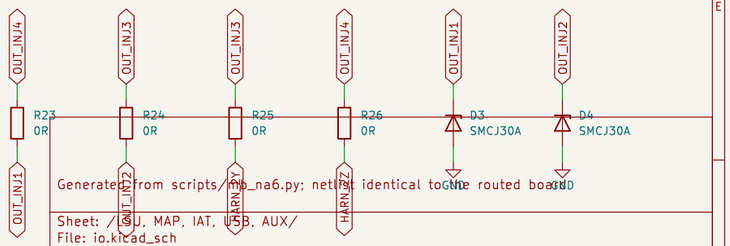
Switch-input 12 V pull-ups, NOT fitted: fit only for a signal that switches to ground

Throttle switch plug (1N, 2L): R27 stock-switch pull-up (NOT fitted); R28 sensor on the stock plug: fit R28 (5 V on 1N), move R29 to R30



1989–93 batch harness: two drivers per injector wire, TVS clamp per wire

Sequential conversion: remove R23/R24, fit R25/R26 (never R26 on an automatic)



Generated from scripts/gen_na6.py; netlist identical to the routed board

Sheet: /ECU, MAP, IAT, USB, AUX/
File: io.kicad_sch

Title: Miata ECU motherboard 1989–1993 – LSU, MAP, IAT, USB, AUX

Size: A3	Date: 2026-10-30	**Rev: 0.11**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4